

Name: _____

Class: _____

ACTIVITY SHEET

Chapter 2: Revision

1 Match each term with the appropriate description using an arrow.

a	chemical change
b	chemical property
c	compound
d	distillate
e	distillation
f	electrostatic attraction
g	element
h	filtrate
i	filtration
j	fractional distillation
k	heterogeneous mixture
l	homogeneous mixture
m	magnetic separation
n	material
o	matter
p	mixture
q	physical change
r	physical property
s	residue
t	separating funnel
u	sieving
v	solute
w	solution
x	solvent
y	substance
z	vaporisation

A	homogeneous mixture
B	change in the composition of a substance, giving a new substance
C	change not involving the chemical composition of a substance
D	determined without changing a substance's chemical composition
E	dissolved substance
F	equipment used for separating immiscible liquids
G	having a non-uniform composition
H	having a uniform composition throughout
I	homogeneous material made up of the same type of particle
J	liquid or solution that passes through filter paper during filtration
K	matter that contains different substances with varying composition
L	physical substance that has mass and occupies space
M	pure substance composed of more than one type of atom
N	pure substance made up of only one type of atom
O	relates to the reactions a substance can undergo
P	separating an undissolved solid from a liquid
Q	separation technique based on attraction in a magnetic field
R	separation technique based on attraction in an electric field
S	solid remaining in the filter paper after filtration
T	substance in which the solute dissolves
U	substance with particular qualities
V	technique for separating a mixture based on particle size
W	technique for separating the solid component of a solution
X	technique for retrieving the liquid component of a solution
Y	technique used to separate liquids with similar boiling points
Z	vapour that is condensed back into a liquid

2 Explain the differences between the terms in each of the items below.

a Physical and chemical properties

b Physical and chemical changes

c An element and a compound

d Homogeneous and heterogeneous mixtures

e A solute and a solvent

f A material and a substance

g Filtrate and distillate

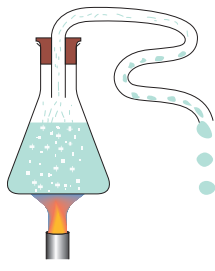
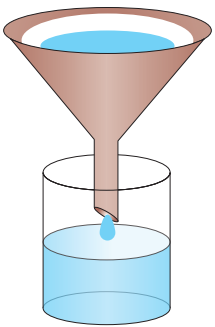
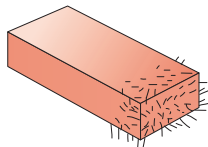
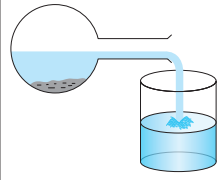
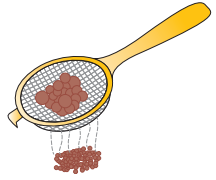
3 Complete the table by identifying an appropriate separation technique and the physical property it utilises.

Mixture	Technique	Physical property utilised
Oil and water		
Water and alcohol		
Molten ions		
Sandy water		
Crude oil		
Iron sand		
Boiling rice		
Salty water		

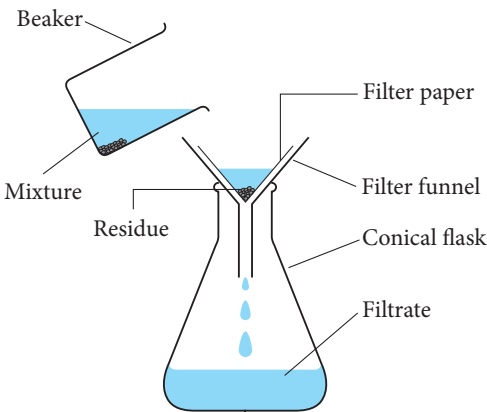
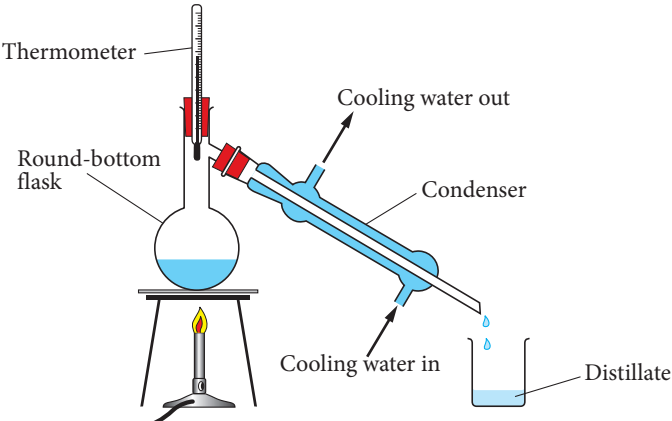
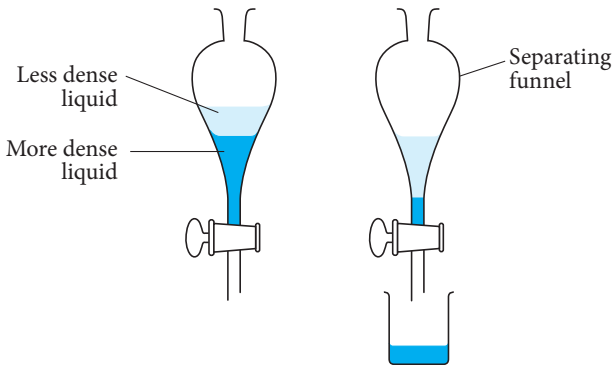
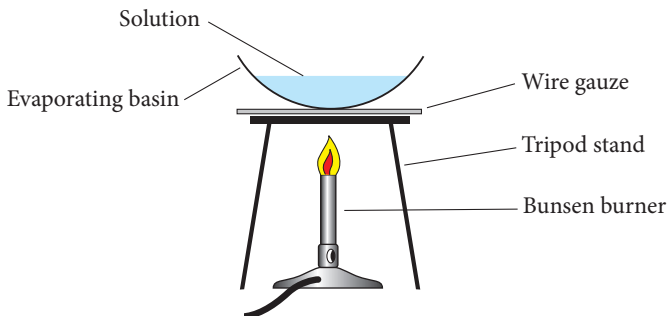
4 Circle items on the list below that are physical properties of substances.

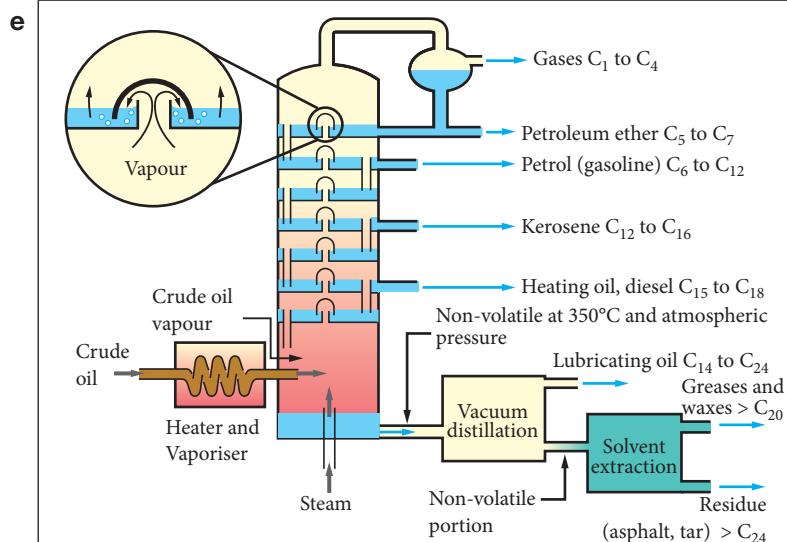
- strength
- reaction with acids
- state
- melting point
- density
- conductivity
- decomposition by heat
- reaction with bases
- colour
- reaction with oxygen
- solubility
- boiling point

5 Identify each of the separation techniques shown below.

6 Explain how the apparatus in each diagram results in the separation of substances in a mixture.

a	 Beaker Mixture Residue Filter paper Filter funnel Conical flask Filtrate	
b	 Thermometer Round-bottom flask Cooling water out Condenser Cooling water in Distillate	
c	 Less dense liquid More dense liquid Separating funnel	
d	 Solution Evaporating basin Wire gauze Tripod stand Bunsen burner	



7 Study this hierarchical classification of materials, then answer the questions below.

a What are the two basic types of materials?

b What are the two types of homogeneous materials?

c What are the two types of pure substances?

